

Jappreet Singh

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Portfolio — GitHub — LinkedIn

Professional Summary

Software Engineer with experience designing and deploying AI-powered applications using Python, FastAPI, Angular, Retrieval-Augmented Generation (RAG), vector databases and Large Language Models. Completed an AI Process & Optimization internship which was focused on AI agent evaluation, workflow automation, and operational improvements. Passionate about enabling organizations to adopt practical AI solutions that streamline business processes, improve productivity, and deliver measurable business value.

Technical Skills

AI & Automation

Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agent Development, Gemini API, Large Language Models (LLMs), Vector Embeddings, Semantic Search, AI Workflow Automation, Supabase Vector Database

Programming Languages

Python, Java, C#, JavaScript, TypeScript, SQL, PHP, Kotlin, HTML5, CSS3

Frameworks

Angular, FastAPI, Spring Boot, .NET, Bootstrap, REST APIs, SOAP APIs

Cloud & DevOps

Azure Fundamentals, Git, Bitbucket, Jira, Vercel, Render, Visual Studio

Databases

MySQL, Supabase, Microsoft Access, Relational Database Design, ER Modeling

Professional Experience

AI Process & Optimization Intern

Cogsdale Corporation

Feb 2025 – Apr 2025

Charlottetown, PE

- Evaluated deployment environments for internal AI agents by designing testing protocols that compared hosting strategies, supporting cost-effective implementation decisions.
- Designed and implemented a beta AI support assistant using no code approach with Copilot studio to automate customer inquiry classification, reducing manual triage effort during pilot testing.
- Analyzed Microsoft GP workflows to identify automation opportunities and documented AI-driven recommendations that improved operational efficiency.
- Collaborated with cross-functional Agile teams using Jira and Bitbucket to manage sprint planning, version control, and peer code reviews.
- Produced over 20 pages of technical documentation covering AI research findings, AI limitations, deployment considerations, and future integration recommendations.
- Researched emerging AI tools and evaluated practical business applications to support internal process optimization initiatives.

Counter Attendant

Tim Hortons

Mar 2024 – Dec 2025

Charlottetown, PE

- Delivered excellent customer service while accurately handling high-volume cash and card transactions in a fast-paced working surrounding.
- Collaborated with team members to maintain operational efficiency and consistently meet service quality standards during peak lunch hours.
- Developed strong communication, multitasking, and problem-solving skills while working under pressure.

Education

Holland College

Computer Information Systems Diploma

2023 – 2025

Charlottetown, PE

- Capstone: Led a three-person Agile team in designing and presenting a complete business and technical proposal for a startup concept, delivering the MVP ahead of schedule.
- Relevant Coursework: Object-Oriented Programming, Data Structures, Systems Analysis, Database Design, Mobile Application Development, Software Engineering, Web Development.

Certifications

- Microsoft Certified: Azure Fundamentals (AZ-900)

Projects

AI Knowledge Assistant with RAG

2025-2026

Angular, FastAPI, Python, Gemini API, Supabase, Vector Search

- Designed and deployed a full-stack AI knowledge assistant enabling document-aware conversations through Retrieval-Augmented Generation (RAG).
- Built an end-to-end document ingestion pipeline supporting PDF uploads, intelligent text chunking, vector embedding generation, semantic indexing, and contextual retrieval.
- Integrated Google's Gemini API with FastAPI to deliver low-latency AI responses using prompt engineering and retrieval-based context injection.
- Implemented streaming responses within the Angular frontend, providing a responsive ChatGPT-style user experience.
- Leveraged Supabase Vector Database to perform semantic search, improving response relevance while reducing hallucinations through retrieval-based prompting.
- Designed RESTful APIs and modular backend services to support scalable AI workflows and future feature expansion.
- Deployed the frontend and backend using Vercel and Render, demonstrating practical business applications of generative AI for knowledge management and workflow efficiency.

Developer Portfolio Website

2025

Angular, TypeScript, GitHub Pages

- Designed and developed a responsive portfolio website showcasing AI projects, software engineering experience, technical skills, and resume.
- Implemented a modern Angular architecture with reusable components, routing, animations, and responsive layouts.
- Integrated GitHub projects and live AI demonstrations to provide recruiters with interactive examples of technical work.
- Optimized the website for performance, accessibility, and mobile responsiveness before deploying through GitHub Pages.

League Tracker Web Application

2024

Java, MySQL, REST, SOAP

- Architected a full-stack sports league management platform supporting team administration, player statistics, scheduling, and standings.
- Developed REST and SOAP APIs enabling communication between backend services and web clients.
- Designed optimized relational database schemas and SQL queries to generate real-time rankings and statistical reports.
- Applied object-oriented programming principles and layered software architecture to improve maintainability.

Employee Management System

2024

C#, PHP, SQL, Bootstrap

- Developed a secure employee management platform supporting authentication, record management, search, and reporting features.
- Implemented session-based authentication, server-side validation, and SQL injection protection to strengthen application security.
- Built dynamic filtering and search capabilities that improved usability when managing employee records.

Home Inventory Application

2024

Angular, TypeScript

- Developed a single-page Angular application using reusable components and modular architecture.
- Implemented CRUD operations, reactive forms, and client-side validation using a simulated JSON backend.
- Applied responsive UI design principles to improve usability across desktop and mobile devices.